No Free Null: Reference Distributions for Explaining Reinforcement Learning Agents

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Abstract. Deciding whether an explanation of a reinforcement learning agent means anything requires a reference distribution: what the explanation would have looked like had the agent learned nothing. Reinforcement learning has no agreed way to build one, and we show the problem is harder than it appears. The construction the field already uses, randomising network parameters, has almost no power: across sixteen checkpoints on four control tasks it never exceeds $z = 2.45$, and its width *equals* the standard deviation of random-initialisation return to three significant figures, so it measures the variance of initialisation rather than anything about the explanation. We then build the three natural replacements and each fails differently. Training on a signal-free corpus varies the corpus as well as the information. Blinding the observation channel removes the agent's capacity to respond, collapsing the reference onto a point mass and firing on a corpus constructed to contain no signal. Permuting the outcomes, the direct analogue of the supervised label-permutation test, leaves the market price forecasting a label it no longer matches and hands the null agents an arbitrage worth \$49 an episode, so the agent under test lands fifteen standard deviations *below* its own null. The pattern is that removing the information always perturbs something else, and the perturbation decides the verdict. We motivate this with a worked case in which an agent with no measurable edge produces an explanation that is stable across seeds, separated beyond its own estimation error, and semantically legible, and with a demonstration that such an explanation can be driven from 39.9% to 3.2% of attribution mass at no measurable cost in return. The choice of reference distribution is the substance of the problem, and it is a choice almost no paper reports making.

1 Introduction

A common argument for interpretability is oversight: if we can see which inputs drive a policy, we can decide whether to trust it. That requires an attribution to mean something, and every attribution method returns an answer whether or not it does. Given any policy and any state, Shapley values [8, 11], integrated gradients [13], and permutation importance all produce a vector of numbers, ranked, signed, and plottable. Nothing in the output distinguishes an attribution reflecting structure the agent exploits from one reflecting nothing at all.

Telling them apart requires a *reference distribution*: what would this explanation have looked like if the agent had learned nothing? In supervised learning there are two standard answers, both from Adebayo et al. [1]. Randomise the model's parameters, or train it on permuted labels; if the explanation does not change, it does not depend on what the model learned. Reinforcement learning has adopted the first and not the second. Huber et al. [7] apply parameter randomisation to Atari saliency maps and observe that the data-randomisation test has no obvious RL analogue: there is no labelled dataset to permute, and it is unclear how a sequential decision maker should respond to artificially corrupted features.

We set out to supply that analogue. The result is mostly negative, and the negative result is the contribution. The construction reinforcement learning already uses has almost no power, for a reason that turns out to be an identity rather than a tendency. Each of the three natural ways to build a better one fails, and each fails differently: removing the information also perturbs something else, and it is the perturbation, not the information, that decides the verdict. *Choosing the reference distribution is the hard part of explaining a reinforcement learning agent, and it is a choice almost no paper reports making.*

Contributions.

1. **The incumbent has no power, and we give the mechanism.** Parameter randomisation never exceeds $z = 2.45$ across sixteen checkpoints on four control tasks, including on a converged CartPole policy scoring 404 of 500, where an environment-level null reaches $z = 117$. Its width *equals* the standard deviation of random-initialisation return to three significant figures, so it measures the variance of initialisation and not the explanation (Section 6.1).
2. **Three replacements, three distinct failures.** A signal-free corpus varies the corpus and widens the reference. Blinding the observation channel removes the agent's capacity to respond and collapses the reference to a point mass, producing a false positive on a corpus built to contain no signal. Permuting the outcomes leaves the price forecasting a label it no longer

matches and hands the null agents an arbitrage worth \$49 an episode (Sections 6.2–6.4).

3. **A worked case of an empty explanation.** On an efficient market an agent with no measurable edge produces attributions that are stable across seeds, separated beyond their own estimation error, and legible. Every conventional check passes (Section 5.1), and a positive control on the same real episodes shows the test separates tasks rather than datasets (Section 5.3).
4. **Attribution is not identified by behaviour.** An auxiliary training penalty drives a feature's attribution from 39.9% to 3.2% of total mass at no measurable cost in return, with no deception and no attack on the explainer. The cost of forging an explanation is the importance of the feature, which is zero exactly where a guarantee would matter (Section 7.1).
5. **An audit of our own choices.** Our thesis implies that our own null must matter, so we checked, found it measuring spans on a different corpus from the observation, and report what that is worth. Six predictions this project made were rejected on its own evidence and are collected rather than dropped (Section 9.1).

What we do not claim. The argument that RL attribution should target outcomes rather than policy outputs is due to Beechey et al. [3], whose SVERL framework unifies behaviour, outcomes and prediction, and whose FastSVERL [2] addresses the scalability, off-policy and temporal problems we only name. Our tests adapt Adebayo et al. [1]. Deletion-based fidelity evaluation is established practice in XRL [5, 9]. We do not offer a null construction that works; we offer an account of why the obvious ones do not, and evidence that the choice changes conclusions.

2 Related work

Shapley values in RL. Beechey et al. [3] give a theoretical account of Shapley-value explanation for RL, showing that earlier applications explained the wrong object and proposing characteristic functions built on agent performance. Their framework identifies three explanatory targets: behaviour, outcomes, and prediction. FastSVERL [2] makes the approach tractable and addresses temporal dependence and off-policy data. We adopt the outcomes framing and implement it independently; we do not extend the theory.

Sanity checks. Adebayo et al. [1] introduced parameter- and data-randomisation tests; Binder et al. [4] examine their limitations. Huber et al. [7] apply the parameter test to perturbation-based saliency for Atari agents and note the data test lacks an RL analogue. We propose one, and show the parameter test is the weaker of the two in this setting.

Manipulating explanations. Slack et al. [12] construct adversarial classifiers that detect the off-manifold points LIME and SHAP query and behave differently on them, yielding innocuous explanations for biased models. That is an attack on the explainer's sampling. Our steering result differs in kind: we train a normal agent with an invariance penalty, the resulting explanation is *correct* about the resulting model, and the point is that behaviour does not pin the explanation down. It is a non-identifiability result rather than an attack.

Uncertainty in Shapley estimation. Goldwasser and Hooker [6] certify top-k Shapley rankings against Monte Carlo error. These guarantees concern the estimator. Section 5.5 shows they are orthogonal to whether the explained model learned anything.

Evaluation in XRL. Surveys note that the absence of ground-truth explanations is a central obstacle, and that perturbation-based fidelity is the common recourse [5, 9]. We use deletion curves in that established sense and do not claim them as a contribution.

3 Method

3.1 The statistic

Let $v(S)$ be the expected episode return of an agent observing only the features in $S \subseteq N$, with features outside S replaced at every timestep by draws from a reference distribution. Define the *attribution span*

$$\Delta \;=\; v(N) - v(\emptyset). \tag{1}$$

Shapley values [11] and integrated gradients [13] satisfy additivity identities (efficiency and completeness respectively) under which per-feature attributions sum to Δ. The span can therefore be read off either.

It is worth stating precisely what Δ is. It is *not* a Shapley quantity: it is the difference between two masked rollouts, and depends only on the masking scheme. Efficiency tells us the Shapley values happen to sum to it. This has a practical consequence we exploit: Δ costs two evaluations rather than $2^{|N|}$.

3.2 The null

The test compares an observed Δ against the distribution of Δ for agents that had nothing to learn. We construct these by corrupting the observation channel (Figure 1).

Blind-observation null. *Replace every observation with an independent draw from a fixed reference distribution matched to the environment's observation moments. Leave dynamics and reward untouched. Train normally.*

Such an agent faces the real task and receives real return; it simply cannot condition on anything. This

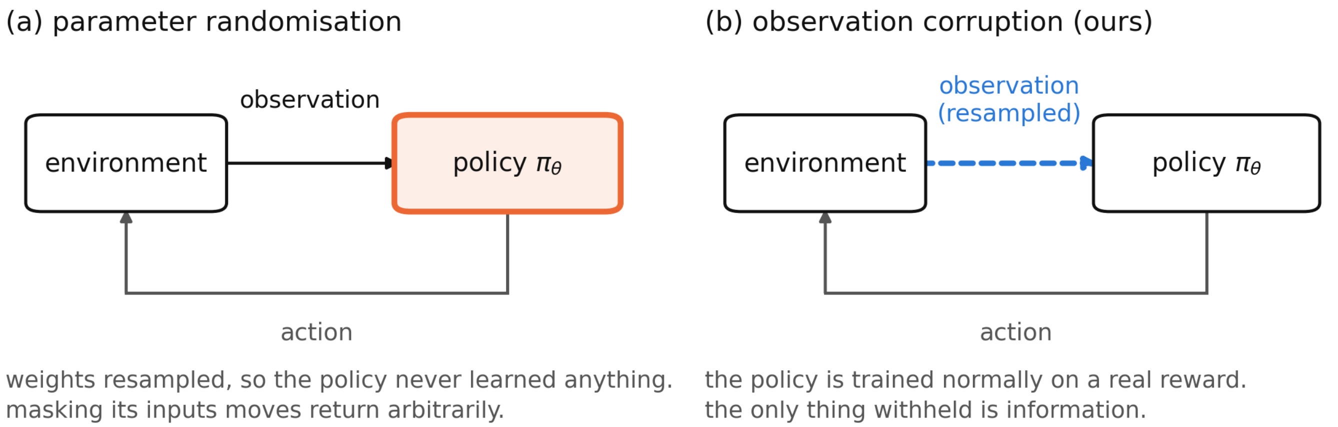

Figure 1: Two ways to build a reference distribution of explanations for an agent that learned nothing. The established check (a) randomises the policy's parameters, leaving a network that was never trained. Ours (b) leaves the policy and its training untouched and corrupts the environment's observation channel, so the agent optimises a real reward with nothing to condition on. Only (b) yields a normally trained policy, which is the failure mode that occurs in deployment; Section 6.1 shows the difference is worth 42× in the width of the resulting reference distribution.

is the RL counterpart of training on permuted labels, and unlike parameter randomisation it yields a *normally trained* policy: the failure mode that occurs in deployment rather than an artificial one.

Matching the reference moments matters. A network fed out-of-range inputs fails for reasons of scale rather than information, which would be a different experiment.

3.3 The test

Given Δ_{obs} and null draws $\Delta_1, \ldots, \Delta_n$ with mean $\bar{\Delta}$ and standard deviation s, we report

$$p_{\text{rank}} = \frac{\#\{i : |\Delta_i - \bar{\Delta}| \geq |\Delta_{\text{obs}} - \bar{\Delta}|\} + 1}{n+1}, \qquad (2)$$

$$z = \frac{\Delta_{\text{obs}} - \bar{\Delta}}{s}, \qquad (3)$$

and require both to reject.

Two statistics are reported because neither suffices. The rank statistic assumes no distributional shape but bottoms out at $1/(n+1)$: with $n = 8$ it cannot reject at $\alpha = 0.05$ however extreme the observation, and in an early run it reported $p = 0.111$ for a signal lying 10.5 standard deviations outside the null. The normal statistic has resolution but assumes a shape the null need not have. Requiring both keeps the test conservative in the direction that matters.

Degenerate nulls. If every null draw is identical, $s = 0$ and z is undefined. We return $z = \pm\infty$ and surface the degeneracy rather than guarding it away, because silently returning $z = 0$ would convert an overwhelming result into a null one.

It is tempting to read a degenerate null as maximally informative, since any deviation from a point mass is infinitely surprising. An earlier version of this paper said exactly that, and Section 6.3 shows it is wrong: a point-mass null has no specificity, and we produce a false positive with one on a corpus that provably contains no signal. Degeneracy is a defect of the construction and a reason to distrust the reference, not a strong result. It occurs whenever a task lets the null agents converge on a single behaviour, which is a property of the null construction rather than of the agent under test.

4 Experimental setup

4.1 Environments with known answers

Explanation evaluation lacks ground truth [5]. We construct three synthetic corpora in which the correct answer is fixed by construction. Episodes are fixed-length with no early termination, so the horizon is controlled exactly.

- **Planted signal.** One of 18 features is informative about the binary outcome; the rest are noise or constants.
- **Null.** Identical shapes and frictions, no informative feature. The optimal policy is to abstain.
- **Decoy.** One feature correlates 0.989 with the outcome in the first 60% of episodes and 0.006 in the remainder.

The claim about the planted corpus needs care. The planted feature is the only one informative about the *outcome*. Others can still affect the *return* without predicting anything, because return depends on behaviour: an agent that cannot tell where it is in an episode acts incoherently and pays transaction costs. The test is that the planted feature ranks first with the majority of attribution mass, not that everything else is zero.

4.2 A market with terminal ground truth

Our main empirical setting is a binary contract settling to \$1.00 if BTC is higher after 15 minutes and \$0.00 otherwise, on a regulated prediction market. We use 6,425 settled contracts over 68 days (89,950 transitions, outcome rate 0.4999), with a walk-forward split by time and purge gaps; the test split is evaluated once.

This instrument was chosen for one property: every episode resolves to a known value, so a critic's $V(s)$ can be checked against realised settlement frequency. Agents are trained with PPO [10]. State is 18 causal features. Actions are target inventory in $\{-1, 0, +1\} \times q_{\max}$. Reward is the change in mark-to-market equity net of costs, which telescopes exactly to net episode P&L.

Transaction costs use the exchange's published taker schedule, $\lceil 0.07\, C\, P(1-P) \rceil$, maximised at $P = 0.5$ where this contract trades by construction. With the measured one-cent modal spread this is a 9% round-trip hurdle on a 50-cent notional, and it determines most of what follows.

4.3 Control tasks

CartPole-v1 (4 features), Acrobot-v1 (6), MountainCar-v0 (2) and Pendulum-v1 (3) provide settings with no relationship to trading and well-understood optimal policies. All four have observation spaces small enough to enumerate every one of the $2^{|N|}$ coalitions, so per-feature Shapley values there are exact. Pendulum's action space is continuous; we discretise it into nine evenly spaced values so the same categorical PPO trains on every task, at the cost of fine control near the optimum.

4.4 Reproducibility

All results derive from public endpoints with no API keys. Hyperparameters were fixed a priori from synthetic sweeps where the correct answer is known analytically, not tuned on validation performance. The full pipeline is one command; 235 tests cover the environment, the attribution implementations, and six no-lookahead properties verified by mutation testing. Every numerical claim in this paper is asserted against its source artifact by a script that runs as the pipeline's final stage.

Code and compute. Code, data-ingest scripts and the exact commands that produce every figure and table are at `https://github.com/Aaryansi/nano-l1`. All experiments run on CPU: an 8-core Apple M1 with 8 GB of memory, no GPU at any stage. The complete pipeline takes roughly two hours from a cold cache, of which about 27 minutes is rate-limited market ingest, and roughly one hour warm. Policies are two-layer MLPs of 64 hidden units; the 18-feature market agent has 5,636 parameters. The headline experiments are therefore reproducible on a laptop, which is a deliberate property of the setting rather than an accident: the claims concern whether an explanation carries information, and that question does not require scale to pose.

Hyperparameters. PPO with clipped surrogate, GAE, and $\gamma = 1.0$; γ is unity because episodes are finite-horizon and reward telescopes exactly to terminal P&L, so discounting would bias the objective away from the quantity of interest. Market agents train for 100 updates over 5 seeds; explanation-time agents for 60 updates; null draws use $n = 24$ for the main test and 12–16 for the environment and robustness experiments. Full values, including network widths, learning rate, clip range and entropy coefficient, are in the configuration files in the repository rather than restated here.

The exchange serves a moving window, so re-running the ingest returns a different set of contracts. A cold rebuild five days after the original produced 6,425 episodes against 6,428, sharing 5,960 tickers; *overlapping episodes are byte-identical on every field.* Ingest is deterministic per contract and what changes is membership. All conclusions reproduced (Section 4.5).

4.5 Reproduction from scratch

All figures above come from a corpus rebuilt after deleting every cached file. An earlier corpus, collected five days before and sharing 93% of its contracts, gave PPO -0.418 ± 0.260 ($p = 0.129$) against -0.595 ± 0.176 ($p = 0.100$), and null-test z of $+12.27/+0.72/-0.15$ against $+12.27/+0.72/+0.23$. Every conclusion is identical on both. Buy-and-hold moved from $+0.068$ to -1.608, confirming its earlier positive figure reflected that period's 52.3% outcome rate rather than an edge.

5 An explanation can be empty

Before arguing that the reference distribution is hard to build, it is worth establishing that one is needed. This section shows an explanation that passes every check a practitioner would apply and carries no information, that the test we use has power and specificity where the answer is known by construction, and that on real episodes it separates tasks by what their observations are worth rather than by whether the corpus is synthetic.

5.1 An explanation that passes every check and says nothing

The market is empirically efficient: its implied probability has a weighted mean absolute calibration error of 0.0172 across ten bins, and spot return correlates 0.476 with the outcome but only 0.016 with the residual after the price is accounted for. No policy beats abstention (Table 1); losses track turnover.

PPO's shortfall is its own. On synthetic data containing no signal by construction the same agent scores -0.47 ± 0.47; here it scores -0.595 ± 0.176. Nothing is left to attribute to the market.

Yet its explanations look entirely reasonable. Across

Table 1: Test split, 1,284 held-out episodes. p-values are paired bootstrap against always-flat on the same episodes, necessary because per-episode P&L has a standard deviation near 50 against differences near 1.

Policy	Mean P&L	Trades/ep	p vs flat
always-flat	$+0.000$	0.00	n/a
buy-and-hold	-1.608	1.00	0.23
PPO (5 seeds)	$\mathbf{-0.595 \pm 0.176}$	1.37	**0.10**
logistic (refit)	-5.729	2.52	<0.0001
mean-reversion	-11.615	4.94	<0.0001
random	-18.532	9.33	<0.0001

Table 2: Null-model test on the market. Null from 24 blind-trained agents, span $+8.19 \pm 5.11$.

Case	Span	z	Verdict
planted signal	$+70.89$	$\mathbf{+12.27}$	informative
null corpus	$+11.86$	$+0.72$	not distinguishable
real market	$\mathbf{+9.37}$	$\mathbf{+0.23}$	**not distinguishable**

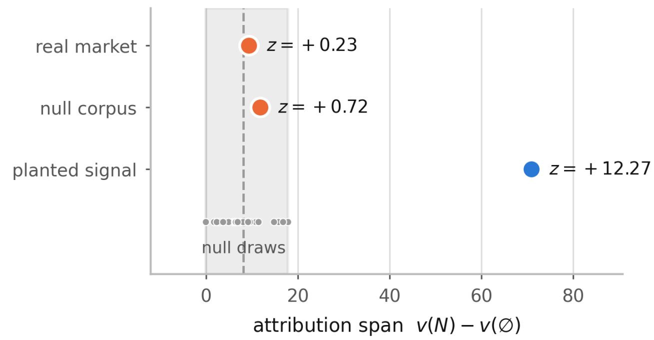

Figure 2: Attribution spans against the null distribution. The shaded band is the range over 24 agents trained where the observation channel carries no information. The market agent sits at its centre.

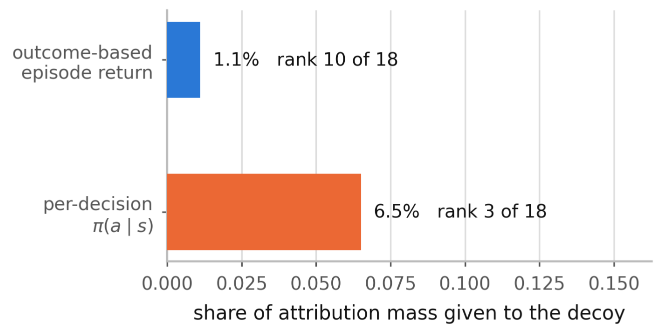

Figure 3: A feature that drives the policy but earns nothing, on held-out episodes where it is provably uninformative.

five seeds whose pairwise performance differences are statistically indistinguishable in all ten pairs: behaviour attributions have cross-seed rank correlation 0.849 (min 0.761) with 100% agreement on the top feature and 100% sign agreement; the top-5 ranking is fully certified, all four adjacent gaps exceeding their combined standard errors; and the ranking is semantically legible, time-to-expiry dominating, which reads as an agent that learned when not to trade.

The null test disagrees (Table 2, Figure 2).

The test has power and specificity. The market agent's explanation sits at the centre of the distribution of explanations of nothing.

We note one consequence for our own reporting. An earlier draft stated that "the whole observation is worth $+5.914$ per episode against being blind" and presented it as a finding. Blind agents produce $+8.19 \pm 5.11$. It was noise.

5.2 Validation against ground truth

On the planted corpus, outcome attribution ranks the planted feature first with 55% of total attribution mass. This check is unavailable on real data and is what licenses the rest.

On the decoy corpus, an agent trained where the decoy predicts the outcome (in-sample $+47.24$, held out -6.83) is explained on held-out episodes where the decoy is provably worthless (Figure 3). Per-decision attribution of $\pi(a \mid s)$ gives it 6.5% of mass, rank 3 of 18; outcome-based attribution gives it 1.1%, rank 10.

Both statements are true. The policy does key on the decoy, so per-decision attribution is correct about behaviour and misleading about value. An auditor asking what an agent relies on that does not work gets the wrong answer from it.

Deletion curves agree: removing features in the order each ranking considers important, the outcome ranking degrades return faster (AUC -187.8 against -165.3; a random ranking gives $+366.6$).

5.3 A positive control on real data

Every case where the test fires elsewhere in this paper is synthetic, which invites the obvious objection: perhaps it detects planted signal rather than information, and would decline on any real corpus. The control holds the corpus, the 18 features, the normalizer, the walk-forward split and the null construction fixed, and varies only the objective.

The market's implied probability is well calibrated, with a weighted mean absolute error of 0.0172. A calibrated price is highly informative *about the outcome* while offering nothing to trade against, because the fee schedule takes more than the edge. So we score the same agent architecture on the same real episodes for *calling* the settlement rather than trading it: $+1$ for each step the call agrees with the eventual outcome, -1 when it disagrees, 0 for abstaining. Reward may reference the settlement; an observation may not, and none does. The environment subclasses the trading one, so the observation construction and normalizer handling are literally the same code and a difference in results cannot be a difference in the environment.

Against a null of 24 agents trained on these same real episodes with the observation channel blinded, the pre-

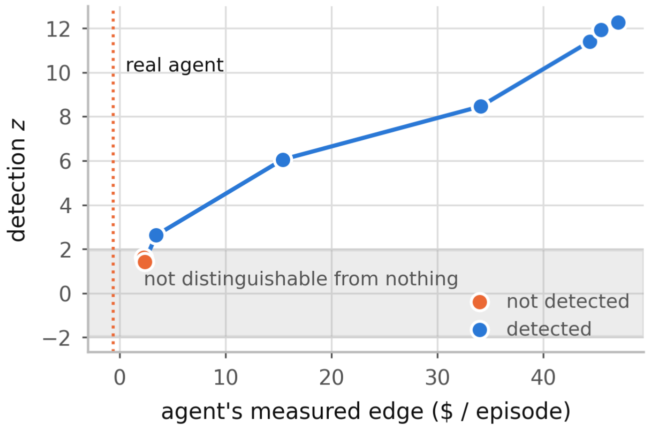

Figure 4: Detection z against the agent's measured edge. The shaded band is where an explanation is not distinguishable from an explanation of nothing.

diction agent's span is $+7.46$ against a null of -0.48 ± 2.34, giving $z = +3.40$: informative, and unchanged in every one of 4,000 resamples. The trading agent on the identical episodes, against the identical null construction, gives $+7.41$ against $+2.47 \pm 2.29$ and $z = +2.16$, which the two-part rule declines. Note that the two spans are almost the same size; what separates them is the null. A blind agent cannot call the outcome at all, so the prediction null sits at zero, while a blind agent can still avoid trading and so retains most of the trading agent's advantage.

The same feature stream therefore carries usable information for one objective and not for the other, and the test says so. This also sharpens what the negative result means. It is not that the market data is uninformative; a calibrated price is very informative. It is that the information is not *monetisable* through a fee schedule that charges 9% round trip, and an attribution of the trading agent cannot tell those two situations apart.

5.4 How much edge is required

Sweeping planted-signal strength calibrates the test (Figure 4). Detection begins near $+3.45$ per episode of *measured* edge. The market agent earns -0.595: an order of magnitude below threshold and of the wrong sign. These edges are in-sample and therefore optimistic; the threshold is a lower bound.

5.5 Estimation guarantees do not substitute

Goldwasser and Hooker [6] certify that a top-k ranking is stable under Monte Carlo noise: a guarantee about the estimator. The market agent's top-5, with standard errors near 0.001 on a leading value of 0.26, has all four adjacent pairs separated beyond their combined error and is fully certified by that criterion. The same explanation fails the null test at $z = +0.23$.

A ranking can be certified stable while the explanation it ranks is indistinguishable from an explanation of nothing. Only one of these two properties is commonly reported.

Table 3: Null construction on four control tasks (CartPole-v1, Acrobot-v1, MountainCar-v0, Pendulum-v1). Entries are mean ± sd of the span across draws; the sd is what sets the test's power.

Environment	Parameter	Environment	Ratio
CartPole	$+55.34 \pm \mathbf{138.92}$	$-1.35 \pm \mathbf{3.29}$	42×
Acrobot	$+64.03 \pm \mathbf{124.94}$	$+0.00 \pm \mathbf{0.00}$	degenerate
MountainCar	$+0.00 \pm \mathbf{0.00}$	$+0.00 \pm \mathbf{0.00}$	0×
Pendulum	$-152.16 \pm \mathbf{131.42}$	$+11.12 \pm \mathbf{37.79}$	3×

6 The reference distribution is the hard part

This is the paper's main section. Every result above was measured against some reference distribution, and the verdicts depend on which one. We show the construction reinforcement learning already uses has almost no power and give the mechanism, then build the three natural replacements in turn. Each fails, and no two fail the same way.

6.1 The established null does not transfer to RL

We construct both nulls for the same task (Table 3, Figure 5). Shapley values here are exact.

A randomly initialised policy behaves arbitrarily, so masking its inputs moves return unpredictably and the reference distribution is very wide. The resulting test has almost no power. Across sixteen checkpoints the parameter null never exceeds $z = 2.45$, including on a converged CartPole policy scoring 404 of 500, while the environment null reaches $z = +117.6$. The two disagree on five of the sixteen. The choice of null is not a detail; it decides the verdict, and the incumbent construction is the one that fails.

Two environments qualify the claim rather than extend it. MountainCar is degenerate under *both* constructions, at 0.00 ± 0.00: within our step budget no policy ever reaches the goal, every episode returns -200, and there is genuinely nothing to explain. The test declines, correctly, and contributes no evidence either way. Pendulum narrows the gap to 3× because its environment null is itself wide (37.79): a blind agent on a dense-reward continuous task still varies in return. The 42× figure is the extreme case, not the typical one.

Acrobot additionally shows the test tracking *whether there is anything to explain* rather than agent quality. Its agent fails completely until it learns (-500 at 10%, 25%, 50% of training), the span is exactly 0.00 there, and the test correctly declines. CartPole behaves differently because its observations matter at every skill level. We do not claim undertrained agents generally produce empty explanations; they do not.

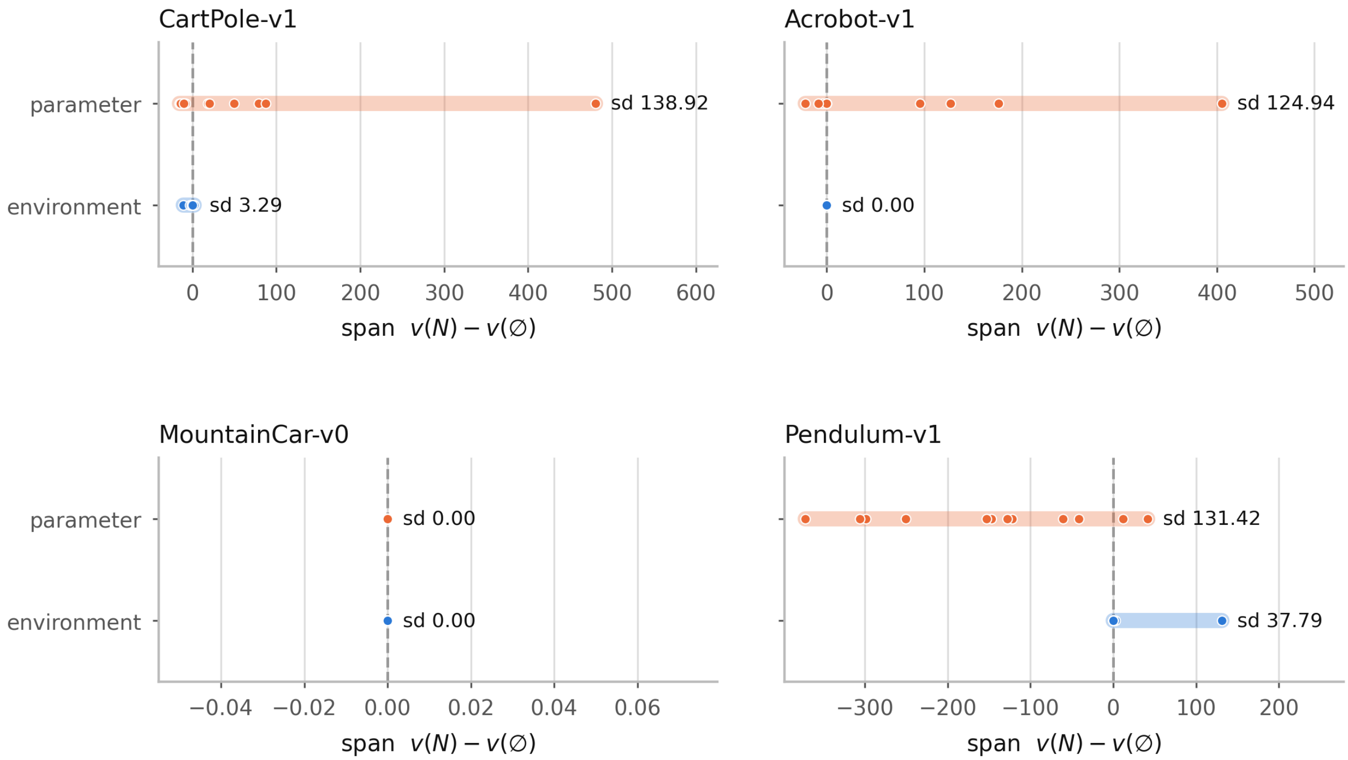

Figure 5: Spread of each null construction on four control tasks. Width determines power: a very wide reference distribution cannot reject anything. MountainCar is degenerate under both constructions, because no policy in the budget ever reaches the goal.

Why the parameter null is wide. The gap has a mechanism, and it is not about explanations at all. Recording each random network's *unmasked* return alongside its span (Figure 6) shows the parameter null's width is, to three significant figures, the spread of random-initialisation return: 138.38 against 138.92 on CartPole, 123.04 against 124.94 on Acrobot, 135.49 against 131.42 on Pendulum, and 0.00 against 0.00 on MountainCar.

The reason is visible in the decomposition $\Delta = v(N) - v(\emptyset)$. For a randomly initialised network the masked term is nearly constant across draws (its standard deviation is 1.8% of the unmasked term's on CartPole and 5.4% on Acrobot) because a policy that was not using its observations loses almost nothing when they are removed. The span therefore inherits the variance of the unmasked term entire. *Parameter randomisation measures the variance of initialisation, not anything about the explanation being tested*, which is why its reference distribution is wide wherever random networks vary and why a test built on it has no power. An environment null does not have this problem because its agents are trained to convergence and land near a common blind-optimal return.

This supersedes the conjecture we offered in an earlier draft, that the width depended on how often a random network is accidentally competent. That is roughly right but imprecise: what matters is not competence but *variance* in competence, and the relationship is an identity rather than a tendency. Four environments is too few to fit anything, and we report the correspondence rather than a correlation coefficient for that reason.

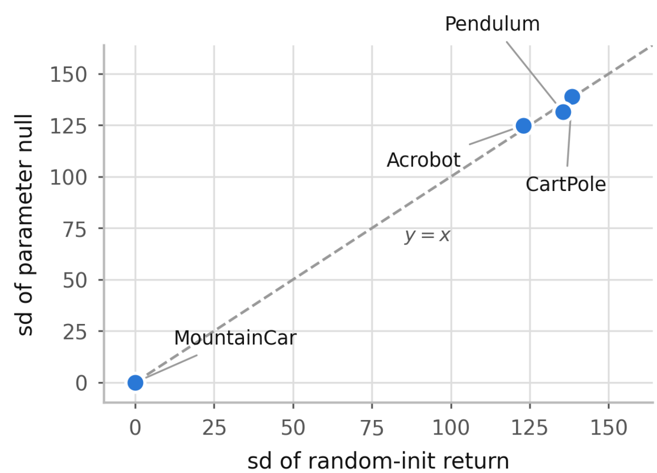

Figure 6: The parameter null's width against the spread of random-initialisation return, one point per environment. The points lie on $y = x$: the established null's reference distribution is the variance of initialisation.

6.2 Our own null construction, checked

The argument above applies to this paper, so we checked. Section 3 defines the null as corrupting the observation

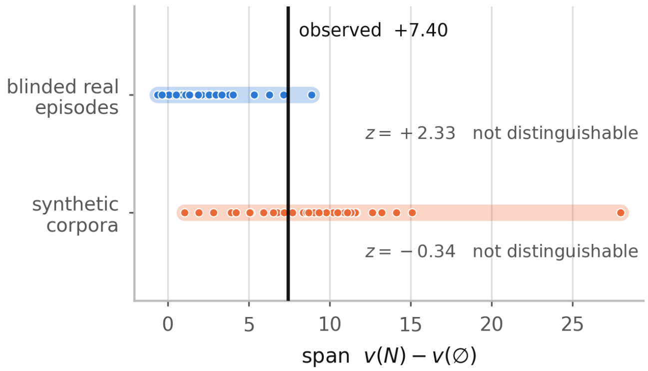

Figure 7: One agent, one observed span, two null constructions. The vertical line does not move; only the reference does. The synthetic-corpus null is nearly twice as wide, because it absorbs corpus-to-corpus variation the observation does not have.

channel while leaving dynamics and reward intact. The null behind our market result does something else: its agents are trained *and* attributed on synthetic signal-free corpora, so their spans are measured on synthetic episodes while the observed span is measured on real ones. The two quantities being compared are not measured on the same thing.

Rebuilding it as defined, with agents trained on the *real* training episodes under a blinded observation channel and attributed on the same real test episodes, and holding the agent and the measurement corpus fixed so the construction is the only variable ($n = 32$ each):

Null	Mean ± sd	z	Verdict
synthetic corpora	$+9.07 \pm 4.91$	-0.34	not distinguishable
blinded real	$+2.26 \pm 2.21$	$+2.33$	not distinguishable

The verdict survives, and how it survives is worth stating: under the blinded-real null the normal statistic *does* reject, the rank statistic does not, and Section 3 requires both. Resampling (Section 7.4) leaves the synthetic-corpus verdict unchanged in 100% of replicates and the blinded-real one in 65%. An earlier run at $n = 12$ put the two constructions on opposite sides of the threshold; that was a small-sample artefact and it disappeared at $n = 32$. We report it because $n = 12$ null agents is a common budget.

We intended at this point to adopt the blinded-real construction throughout. The next three sections are why we did not.

6.3 Correcting the null everywhere, and why we did not

Section 6.2 argues that a null should hold the corpus fixed and vary only the information. Taken seriously that is a per-case construction: for each case, train the null agents on *that* case's episodes with the observation channel blinded, and attribute them on the same rollout as the observation. We built it and ran all three cases. It cannot be used.

Case	Span	Null	z	Verdict
planted signal	$+74.69$	0.00 ± 0.00	$+\infty$	fires
null corpus	$\mathbf{+13.90}$	0.00 ± 0.00	$+\infty$	**fires**
real market	$+7.40$	2.97 ± 2.29	$+1.93$	declines

The middle row is a corpus constructed to contain no informative feature. The test fires on it. The construction has power and no specificity, and a test with no specificity is not a test.

The two nulls are not the same null. The mechanism explains the original design better than the original design explained itself. A *signal-free corpus* gives agents that can see, in an environment whose observations carry no information about the outcome; they still respond to observational structure, so their spans are nonzero and spread out. A *blinded channel* gives agents that cannot see; they converge to a constant policy, and a constant policy's return does not depend on its inputs at all, so every null span is exactly zero.

The agent under test is sighted. The correct reference is therefore sighted agents with nothing to learn, not blind ones. Blinding removes the agent's capacity to respond to observational structure along with the information, and the null collapses onto a point mass at zero against which any nonzero span is declared significant. This is why the degeneracy noted in Section 3 is a defect rather than a strong result, and why the Acrobot degeneracy in Section 6.1 should be read as a warning about the reference rather than as a curiosity.

An independent horizon sweep found the same collapse before we understood it. Extending the episode to 56 steps drives the blinded null to a standard deviation of 0.04 with two distinct values across sixteen agents: every blind agent converged on the same abstaining policy. That is the Acrobot degeneracy again, and the mechanism is the one above rather than anything about horizon.

The collapse is a matter of convergence. On the synthetic corpora the null is a point mass; on the real market it has width 2.29. That difference is not about the corpora but about how far the blind agents got. The real training split is roughly four times the size of the synthetic ones, so the same 40 updates is a quarter as many passes over the data. Sweeping the null agents' budget with everything else fixed (Figure 8) shows the null closing on zero as their return approaches abstention, and the verdict on the real market changing with it:

Updates	Null mean ± sd	z	Blind return	Verdict
20	$+5.60 \pm 3.27$	$+0.55$	-8.74	declines
40	$+3.55 \pm 2.25$	$+1.71$	-4.13	declines
80	$+0.55 \pm 1.62$	$\mathbf{+4.23}$	-0.18	**fires**
160	-0.02 ± 0.04	$\mathbf{+203.6}$	-0.05	**fires**

So the construction does not merely fail on synthetic corpora. Trained to convergence it reports the market

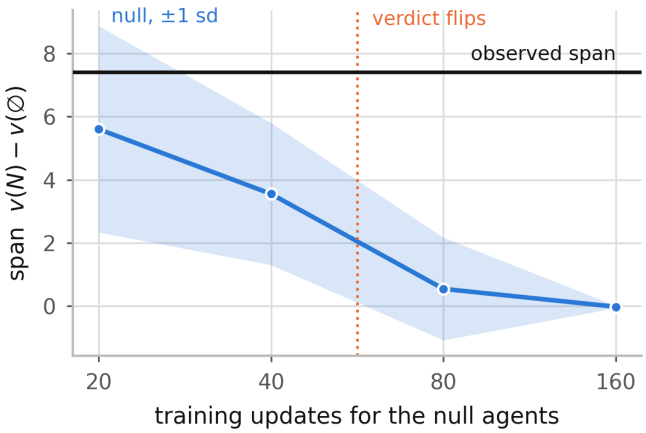

Figure 8: The blinded null collapses as its agents converge on abstention. The observed span is fixed; only the reference moves. A null that tightens with its own training budget cannot be read as a property of the agent under test.

agent's explanation as informative at $z = +203.6$, which is the opposite of what every other measurement in this paper says, and it does so for a reason that has nothing to do with the agent: the null's standard deviation has fallen from 3.27 to 0.04 while the observed span has not moved at all. A reference distribution that tightens with its own training budget cannot be read as a property of the agent under test, and a test whose verdict is set by that budget is measuring the budget.

6.4 A third construction, and a pattern

Section 6.3 leaves an obvious gap: hold the corpus fixed *and* keep the null agents sighted. There is a natural way to do it, and it is the construction this paper claimed to be supplying all along. Permute the outcomes across episodes and leave everything else alone. The observation stream is untouched, because the feature block is a function of the quote and spot arrays and never reads the settlement, so no observation moves by a single bit. The marginal outcome rate is preserved exactly, since a relabelling is not a resample. This is the direct analogue of the label-permutation test of Adebayo et al. [1].

It has power and specificity (Figure 9): on the planted signal it fires at $z = +10.48$, and on the signal-free corpus it declines at $z = +1.75$. No null collapses. It is nonetheless unusable, and the way it fails is the most instructive of the three.

Case	Span	Null	z	Verdict
planted signal	+74.69	$+11.89 \pm 5.99$	+10.48	fires
null corpus	+13.90	$+7.96 \pm 3.39$	+1.75	declines
real market	**+7.40**	$\mathbf{+55.85 \pm 3.12}$	**−15.52**	**below null**

The market agent's span is not merely inside the reference distribution, it is fifteen standard deviations *below* it. The null agents earn +36.26 per episode where the observed agent earns −1.13. They are not agents with nothing to learn; they are agents with a great deal to learn.

Permuting a label that the observations forecast. The price is a forecast of the outcome. Permuting the outcome leaves the forecast in place and attaches it to a different contract, so the market becomes grossly mispriced: a contract whose price has moved to 0.99 now settles at 0.49. Fading the terminal price, a rule needing nothing but the quote, earns −0.005 per contract in the real corpus and +0.486 in the permuted one, which at the position cap is roughly $49 an episode. Calibration error rises from 0.007 to 0.456.[1] The null agents learn that arbitrage, it requires seeing the price, and their spans inflate accordingly.

This is not a quirk of prediction markets. *Wherever an observation is a forecast of the label, permuting the label makes that forecast exploitably wrong*, and the null agents are handed a task easier than the real one rather than harder. Supervised learning does not have this problem because the model does not act on its input and cannot trade against it.

The pattern. Three constructions, three failures, and the same shape each time. Removing the information also perturbs something else, and the perturbation decides the verdict:

Construction	Also changes	Consequence
signal-free corpus	the corpus	widens the reference; errs toward declining
blinded channel	the agent's capacity to respond	collapses to a point mass; fires on anything, budget-dependent
outcome permutation	the price's calibration	creates an arbitrage; the null learns more than the agent

We report this as the finding rather than presenting a fourth construction we have not tested. A null for reinforcement learning has to remove the information while leaving the observation stream, the agent's responsiveness, and the task's difficulty intact, and none of the three natural ways of doing that manages all four. Huber et al. [7] observed that the data-randomisation test has no obvious RL analogue. We set out to supply one and instead have an account of why it is hard.

What we keep. The results in this paper use the signal-free-corpus null, whose fault is the mildest of the three: it widens the reference and so errs toward declining, which is the conservative direction for every claim we make with it. That is a reason to trust our negative results more than our positive ones, and we would not have been able to say which way the bias ran without building the other two. One construction we have not tried is a *stratified* permutation, shuffling outcomes only

[1] Terminal mid on the training split in eight quantile bins, which is not the estimator behind the figure in Section 4.2; only the contrast is meant to be read.

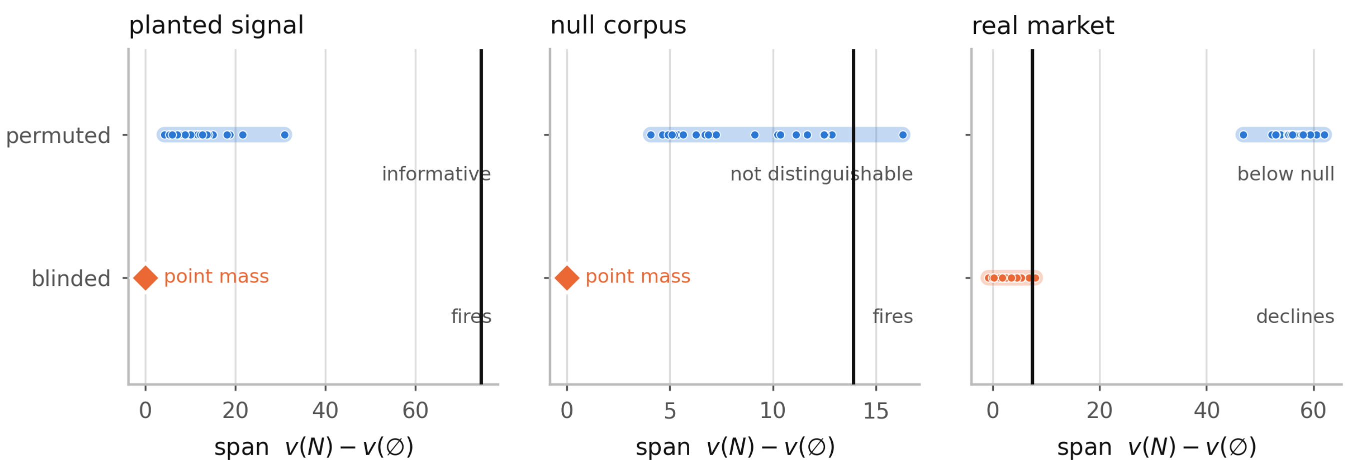

Figure 9: Two null constructions on the same three cases. The vertical line is the observed span and does not move; only the reference does. Blinding the observation channel yields a point mass on both synthetic corpora, which is why it fires on the one built to contain no signal. Permuting the outcomes yields a usable reference everywhere, and on the real market places it so far above the observed span that the agent lands fifteen standard deviations below its own null.

Table 4: Steering one feature's attribution.

Corpus	Attribution	Return	p
market	39.9% → **3.2%**	−3.49 → −1.65	**0.66**
planted signal	46.5% → **1.5%**	+45.04 → **+7.92**	<0.001

within price buckets, which would preserve calibration and remove only the information beyond the price. It asks a narrower question, whether the agent knows anything the market does not, and that may be the better question.

7 What survives the choice of null

Not everything in this paper depends on the null. Three results do not: they compare an agent against itself or against a second attribution method, so the reference distribution never enters. They are the results we would defend hardest.

7.1 Attribution is not identified by behaviour

We add an auxiliary penalty during training on the divergence between $\pi(\cdot \mid s)$ and $\pi(\cdot \mid s')$, where s' has one feature resampled from the batch marginal: the same interventional perturbation the attribution measures (Table 4, Figure 10).

The attribution is equally suppressible in both cases (92% and 97%). What differs is the price: where the feature is the planted signal, suppressing it destroys 81% of return; on the market it costs nothing measurable. Explanations are steerable at no cost exactly where they are not tracking anything, corroborating Section 5.1 by a route sharing no machinery with the null test. The control matters: had steering succeeded on both corpora, the penalty would merely be defeating the attribution method.

7.2 What the verdict does and does not depend on

Attribution family. Integrated gradients [13], sharing no machinery with Shapley, ranks features at 0.981 correlation with Shapley across five seeds and is comparably seed-stable (0.750 against 0.800). Per-feature attributions are not a Shapley artefact. IG cannot test the outcome-level claim: episode return is not differentiable through the environment, so IG has no outcome analogue. Its behaviour-level span reports the market agent as informative ($z = +4.58$), which is consistent rather than contradictory and is precisely the behaviour-versus-outcome distinction of Beechey et al. [3]. The agent's behaviour does respond to its observations; that responsiveness earns nothing.

Credit assignment. Leave-one-out ($v(N) - v(N \setminus i)$) and only-one-in ($v(\{i\}) - v(\emptyset)$) are the extremes of the average Shapley takes, and neither satisfies efficiency, so each carries its own total. On the planted signal all three fire ($z = +10.66$, $+10.96$, $+2.29$); on the market none does ($z = -0.42$, -1.57, -0.72). The verdict is scheme-independent. *The per-feature ranking is not*: leave-one-out and only-one-in rank features at only 0.309 correlation with each other on the same agent. Whether there is anything to explain is robust; which features matter is not, and a ranking from a single scheme should not be reported as the answer.

7.3 Off-manifold masking

The standard objection to perturbation attribution applies here and deserves a direct answer. We define $v(S)$ by replacing the features outside S with draws that ignore the features inside S. Our 18 market features are

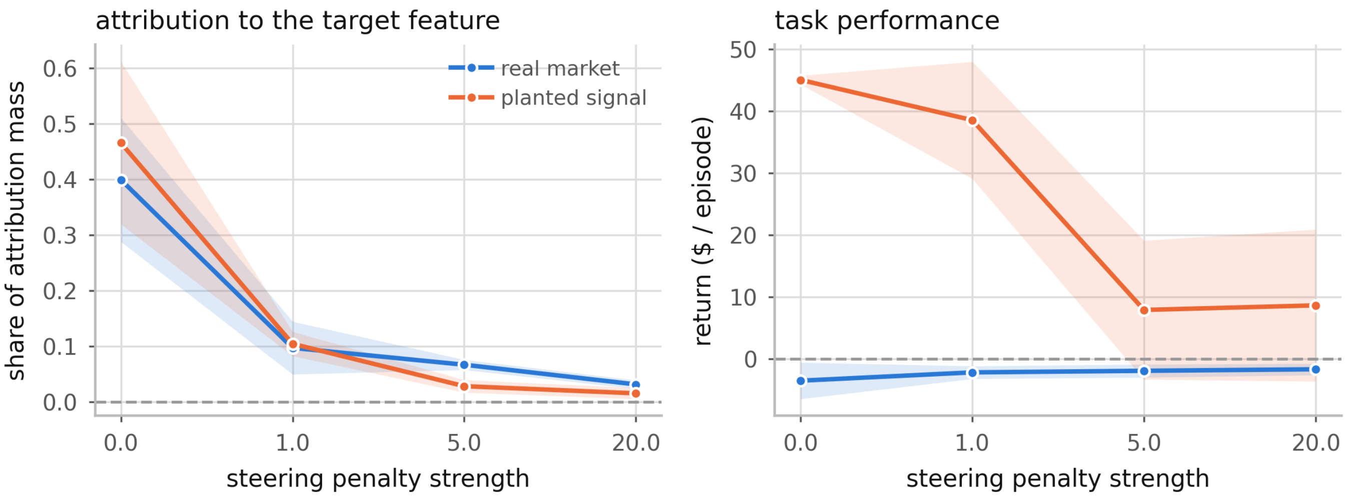

Figure 10: Attribution is equally suppressible on both corpora (left). Only where the feature is load-bearing does suppressing it cost return (right).

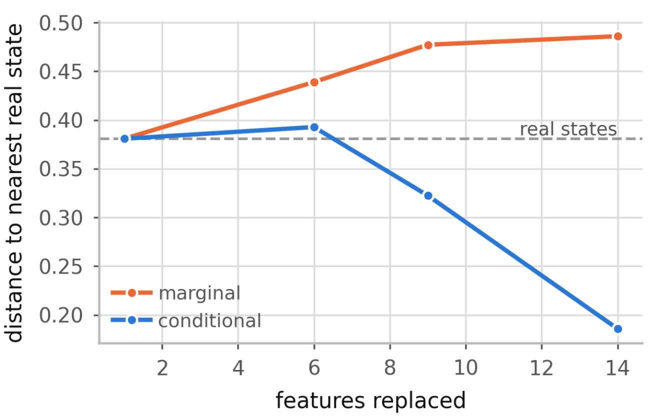

Figure 11: Distance from a masked state to the nearest real state, in per-feature standard deviations, against the number of features replaced. The dashed line is where real held-out states sit relative to the reference sample. Marginal masking moves *above* that floor as more features are replaced; neighbour-conditioned masking stays at or below it. The fully-masked point is omitted: those states are drawn from the reference sample itself, so their distance to it is zero by construction rather than by evidence.

correlated, so those draws break the joint distribution and the policy is scored on states that never occur. Slack et al. [12] build their attack on precisely that gap. If the span reflected distribution shift rather than information, the headline result would be an artefact of the masking.

The span cannot be an off-manifold artefact. This is structural rather than empirical, and we state it as such. The statistic is $\Delta = v(N) - v(\emptyset)$, and neither coalition evaluates an off-manifold state. $v(N)$ masks nothing. At $v(\emptyset)$ our implementation replaces the entire observation from a *single* reference row, so the synthetic state is itself a real observation, merely a different one. Conditioning is vacuous on an empty kept set, since $p(x \mid \emptyset) = p(x)$. A conditional masker must therefore agree with the marginal one on Δ exactly, and ours does: +8.838 under both. That equality confirms the implementation respects the argument; it is not independent evidence for it.

The concern is real in the interior. It would be convenient but false to stop there, because the argument above says nothing about intermediate coalitions, which is where per-feature attributions live. We implemented a conditional masker that draws replacements from the $k = 16$ reference rows nearest the current state in the kept features only, approximating $p(x_{\bar{S}} \mid x_S)$. Measuring the distance from masked states to the nearest real state (Figure 11) shows the two modes are not equivalent: with 14 of 18 features replaced, marginal masking sits at 0.486 against a real-state floor of 0.381, while conditional masking sits at 0.186. Marginal masking does leave the manifold, by a measurable amount, in the interior.

Consequences. Re-deriving the verdict entirely under conditional masking, against a null of 12 conditionally-masked blind agents, reproduces it: the planted signal fires at $z = +9.61$ and the real market does not at $z = -0.55$. The per-feature ranking is another matter. Leave-one-out values under the two modes correlate at only 0.767 by rank, and *they disagree on which feature is most important.* This is the same split Section 7.2 found for credit assignment: whether there is anything to explain is robust, which feature to name is not. A practitioner may report the verdict; the ranking should carry the masking mode with it.

7.4 Confidence intervals on the verdicts

Every z in this paper divides by a standard deviation estimated from 12 to 32 null agents. Reporting it to two decimals hides that. We resample the null draws with replacement and re-apply the full two-part decision rule to each replicate, which gives a percentile interval for z and, more usefully, the fraction of replicates in which the verdict is unchanged.

Every verdict in the paper is stable in 100% of replicates except the two blinded-real market comparisons, at 64% and 65%. The planted signal's interval is $[+10.3, +16.9]$ and the market's, under the null used in Section 5.1, is $[-0.16, +0.73]$: nowhere near the threshold in either direction. This bootstraps the null only. Uncertainty in the observed statistic, a mean over held-out episodes, is not included, so these intervals are optimistic in a way we would rather state than hide.

8 Implications for oversight

The case for interpretability as an oversight mechanism is that seeing which inputs drive a policy tells us something about whether to trust it. Three of our results bear on that argument, and none of them requires an adversary.

Passing every check is not evidence. On a corpus where the correct answer is known to be "nothing", an agent's attribution is stable across seeds, separated beyond its own estimation error, and semantically legible. Every property a practitioner would check is present and the explanation is empty. Stability in particular is worth naming: it is the most commonly reported evidence that an explanation is trustworthy, and Section 9.1 records that we expected instability to be the tell and were wrong. Emptiness and instability are unrelated.

Explanations are cheap to forge, without deception. Section 7.1 drives a feature's attribution from 39.9% to 3.2% of total mass at no measurable cost in return. This needs no deceptive policy, no attack on the explainer, and no knowledge of who will audit the model: an auxiliary training term is enough, and the resulting explanation is *correct* about the resulting agent. The cost of forgery is exactly the importance of the feature, which is zero for features that do not matter. An oversight regime that reads attributions as evidence of what a model relies on is therefore reading a quantity the developer can set, in the region where it would most want a guarantee.

The verdict depends on a choice nobody reports. Section 6.1 shows the established sanity check has almost no power in RL. Section 6.2 shows our own headline is sensitive to the construction, and Section 6.3 shows that the alternative which looks more principled is broken, in a way visible only by running it rather than by reasoning about it. A reported attribution is not interpretable without the reference distribution it was measured against, and that reference is currently neither standardised nor, in most papers, stated.

What we would ask for. Report the null construction alongside any attribution offered as evidence. Treat a per-feature ranking as a joint property of the attribution scheme and the masking distribution, since Sections 7.2 and 7.3 show it moves under both, to the point of disagreeing on the most important feature. The aggregate verdict, whether there is anything to explain at all, is the part that survives; it is also the weaker claim, and it is the one we would trust.

9 What did not survive

A reader assessing whether the surviving claims were stress-tested should be able to see what did not survive, without reconstructing it from the sections that happen to mention it.

9.1 Hypotheses we rejected

Six predictions in this project turned out to be wrong (Table 5). They are collected rather than left implicit in the sections that tested them, because a reader assessing whether the surviving claims were stress-tested should be able to see what did not survive.

One further claim is stated in the paper as structural rather than empirical. The span's invariance to the masking mode (Section 7.3) follows from the construction, since $p(x \mid \emptyset) = p(x)$; the measured equality confirms the implementation respects that argument and is not independent evidence for it. We separate the two because a numerical agreement that must hold is easily mistaken for one that might not have.

10 Limitations

The construction is adapted. The test is the data-randomisation test of Adebayo et al. [1] with an RL-appropriate null; the outcomes framing is Beechey et al. [3]'s; deletion-based fidelity is standard.

The span measures information in aggregate. An explanation could clear the test and still misattribute among features. Section 7.2 shows the ranking is scheme-dependent even when the verdict is not, and Section 7.3 shows it is masking-dependent too, to the point of disagreeing on the most important feature. Only the verdict survives both.

Conditional masking is approximated, not solved. Our conditional masker is k-nearest-neighbour in the kept features, which is a practical proxy for $p(x_{\bar{S}} \mid x_S)$ and inherits the usual difficulties in high dimension. It is adequate to show the two modes differ; it is not a claim to have sampled the true conditional. Neither mode fixes the deeper issue that replacements are redrawn each timestep, so masked *states* are plausible while masked *trajectories* are not.

Temporal credit assignment is named, not solved. Masking a feature for a whole episode measures whether it mattered, not when. This is FastSVERL's [2] territory.

Table 5: Predictions this project made and then rejected on its own evidence. None of the surviving results depends on any of them.

Hypothesis	Prediction	Outcome
An agent with no edge produces unstable explanations, and instability is the tell.	Low cross-seed rank correlation.	**Rejected.** 0.850 mean rank correlation, unanimous on the top feature. Emptiness and instability are unrelated, which is what makes the empty case hard to catch.
Outcome-level attribution, being tied to return, is more stable than behaviour-level attribution.	Higher cross-seed rank correlation for outcomes.	**Rejected, and reversed.** Outcomes reach 0.347 against behaviour's 0.850. Attributing to return is noisier, not steadier.
The parameter null is wide because randomising weights perturbs the measurement's domain and the error compounds along the trajectory.	Its variance grows with horizon faster than the environment null's.	**Rejected.** Over horizons 4 to 28 both grow at indistinguishable rates ($\alpha = 0.94$ against 0.83), ratio flat at 1.7–1.8×.
The parameter null is wide because a randomly initialised network is sometimes an accidentally competent policy.	Width tracks how often random policies do well.	**Superseded.** Roughly right but imprecise: the width is not related to competence but *equal* to the standard deviation of initialisation return (Section 6.1). A tendency was replaced by an identity.
A null that holds the corpus fixed, by blinding the observation channel, is more principled and should replace the shared signal-free-corpus null.	The same verdicts against a tighter reference, and more power.	**Rejected.** It fires on a corpus built to contain no signal, and its verdict on the real market flips with the null agents' training budget (Section 6.3). Blinding removes the capacity to respond to observational structure, not only the information.
Permuting the outcomes fixes both faults: the corpus is held fixed and the null agents stay sighted.	Power, specificity, and a usable reference on all three cases.	**Rejected.** It has power and specificity and still fails. The price forecasts the outcome, so relabelling makes it exploitably wrong: the null agents earn +36.26 against the observed agent's −1.13 and the market's span lands 15 sd *below* its own null (Section 6.4).

Off-policy attribution is not handled. Masked coalitions induce a different policy and therefore a different state distribution; estimates are on-policy for the masked policy, not counterfactuals about the unmasked one.

The width identity is measured on four environments. The correspondence between the parameter null's width and initialisation-return variance is close to exact on each task we ran, and it has a mechanism, but four points cannot establish a law. We report the correspondence, not a fitted relationship.

The null construction is chosen, not derived. We keep the signal-free-corpus null because the corpus-matched alternative fails (Section 6.3), not because we can show it is optimal. Its known fault is that it varies the corpus as well as the information, which widens the reference and biases toward declining. A construction that holds the corpus fixed *and* keeps the null agents sighted would dominate both, and we do not have one.

Scope. One real domain and four classic-control tasks; one attribution family for the outcome-level tests. All four control tasks have small observation spaces, so nothing here speaks to pixel-based or high-dimensional agents, where the reference distribution is itself the hard problem. Box2D environments were deliberately excluded: they require a system-level build dependency, which would trade the one-command reproduction for one more observation dimension. Corpus membership is not reproducible across time, though content is.

11 Conclusion

An attribution of a reinforcement learning agent is uninterpretable without the reference distribution it was measured against, and reinforcement learning does not have one it can rely on. The construction the field uses measures the variance of initialisation. Of the three natural replacements, one varies the corpus, one removes the agent's capacity to respond and fires on a corpus built to contain no signal, and one hands the null agents an arbitrage. In each case removing the information perturbed something else, and the perturbation decided the verdict.

We think the general shape is now visible. A usable null has to remove the information while leaving the observation stream, the agent's responsiveness, and the difficulty of the task intact. The three constructions here each sacrifice exactly one of those, which is why each fails in its own direction and why the failures were not predictable from first principles. We found them by building all three.

That matters because attributions are being offered as evidence. An explanation can pass every conventional check and be empty, and it can be driven to say something else at a price equal to the importance of the feature being hidden, which is zero exactly where a guarantee would be worth having. Neither of those is detectable without a reference distribution, and choosing one is not a detail of the method. It is the method.

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